

$$1.2. (1) (11101011)_2$$

$$= (EB)_{16}$$

$$(2) (1010110101)_2$$

$$= (2B5)_{16}$$

$$(3) (11100101110)_2$$

$$= (72E)_{16}$$

$$1-4. (1) (4AC)_{16}$$

$$= (10010101100)_2$$

$$(2) (AB9)_{16}$$

$$= (1010110010111001)_2$$

$$(3) (78ADF)_{16}$$

$$= (0111100010101101111)_2$$

$$(4) (98EBC)_{16}$$

$$= (10011000111010111100)_2$$

$$1-7. (1) (100)_{10}$$

$$= (1100100)_2$$

$$= (64)_{16}$$

$$(2) (10)_{10}$$

$$= (1010)_2$$

$$= (A)_{16}$$

$$(3) (110)_{10}$$

$$= (1101110)_2$$

$$= (6E)_{16}$$

$$(4) (88)_{10}$$

$$= (10110000)_2$$

$$= (58)_{16}$$

$$1-8. (987)_{10}$$

$$= (10011000111)_{8421}$$

$$(3456)_{10}$$

$$= (0011010001010110)_{8421}$$

$$(7531)_{10}$$

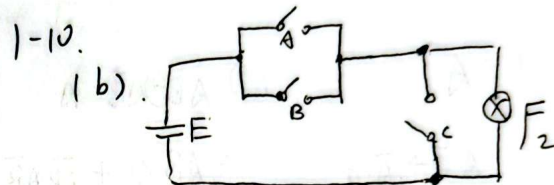
$$= (0111010100110001)_{8421}$$

$$1-9. (010110001001)_{8421}$$

$$= (589)_{10}$$

$$(1000100100111000)_{8421}$$

$$= (8938)_{10}$$



A	B	C	F ₂
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0

$$\bar{A}B\bar{C} + A\bar{B}\bar{C} + ABC$$

$$= \bar{A}B\bar{C} + A\bar{C}$$

$$= \cancel{AC} + \cancel{AB}$$

$$= A\bar{C} + B\bar{C}$$



1-11.

(1) 若 $A+B=A$, 则 $B=0$.
正确.

(2) $1+B=AB$, 则 $A=B=1$

$$\Rightarrow AB=1.$$

$$\Rightarrow A=1 \text{ 且 } B=1.$$

正确.

(3) 若 $AB=AC$, 则 $B=C$.

错误.

当 $A=0$ 时, 恒成立.

1-19. (1) $F(A,B,C) = \sum(m_0, m_1, m_2, m_4, m_5, m_7)$

	$\bar{B}\bar{C}$	$\bar{B}C$	$B\bar{C}$	BC
$\bar{A}$	1	1	0	0
A	0	0	1	1

$$\bar{B} + AC + \bar{A}\bar{C}$$

(3)

AB \ CD	00	01	11	10
00	1	1	1	1
01	1	0	0	1
11	1	0	0	1
10	1	1	1	1

$$\bar{B} + \bar{D}.$$

1-15.

(1) $A \oplus 0 = A$

$$\begin{aligned} A \oplus 0 &= A \cdot \bar{0} + \bar{A} \cdot 0 \\ &= A \cdot 1 + 0 \\ &= A. \end{aligned}$$

(6) $A \oplus \bar{A} = A$

$$\begin{aligned} &= A\bar{A} + \bar{A}A \\ &= A(\bar{A}+A) + (\bar{A}+A)A \\ &= AB + \bar{A}\bar{B} \\ &= A(B+\bar{B}) \\ &= A. \end{aligned}$$

(2) $A \oplus 1 = \bar{A}$

$$\begin{aligned} A \oplus 1 &= A \cdot \bar{1} + \bar{A} \cdot 1 \\ &= A \cdot 0 + \bar{A} \\ &= 0 + \bar{A} \\ &= \bar{A} \end{aligned}$$

(7) $A \oplus \bar{B} = \overline{A \oplus B}$

$$\begin{aligned} A \oplus \bar{B} &= A \oplus B \oplus 1 \\ &= (A \oplus B) \oplus 1 \\ &= \overline{A \oplus B} \end{aligned}$$

(3) $A \oplus A = 0$

$$\begin{aligned} A \oplus A &= A\bar{A} + \bar{A}A \\ &= A\bar{A} \\ &= 0 \end{aligned}$$

(5) $A \oplus \bar{A} = 1$

$$\begin{aligned} A \oplus \bar{A} &= A\bar{A} + \bar{A}A \\ &= A\bar{A} + \bar{A}A \\ &= A + \bar{A} \\ &= 1 \end{aligned}$$



1-20.

$$(1) F = \bar{A}\bar{B}\bar{C}\bar{D} + \bar{A}BC\bar{D} + A\bar{B}\bar{C}D + ABCD + A\bar{B}C\bar{D}$$

$$= (0100)_2 + (0110)_2 + (1101)_2 + (1111)_2 + (1010)_2$$

$$d = \bar{A}\bar{B}\bar{C}\bar{D} + \bar{A}\bar{B}C\bar{D} + \bar{A}B\bar{C}D + \bar{A}BCD + A\bar{B}C\bar{D}$$

$$= (0000)_2 + (0010)_2 + (0101)_2 + (0111)_2 + (1000)_2$$

AB \ CD	00	01	11	10
00	(X)	0	X	(X)
01	1	X	0	1
11	0	1	1	0
10	(X)	0	0	1

$$\bar{A}BD + \bar{A}\bar{D} + \bar{B}\bar{D}$$

